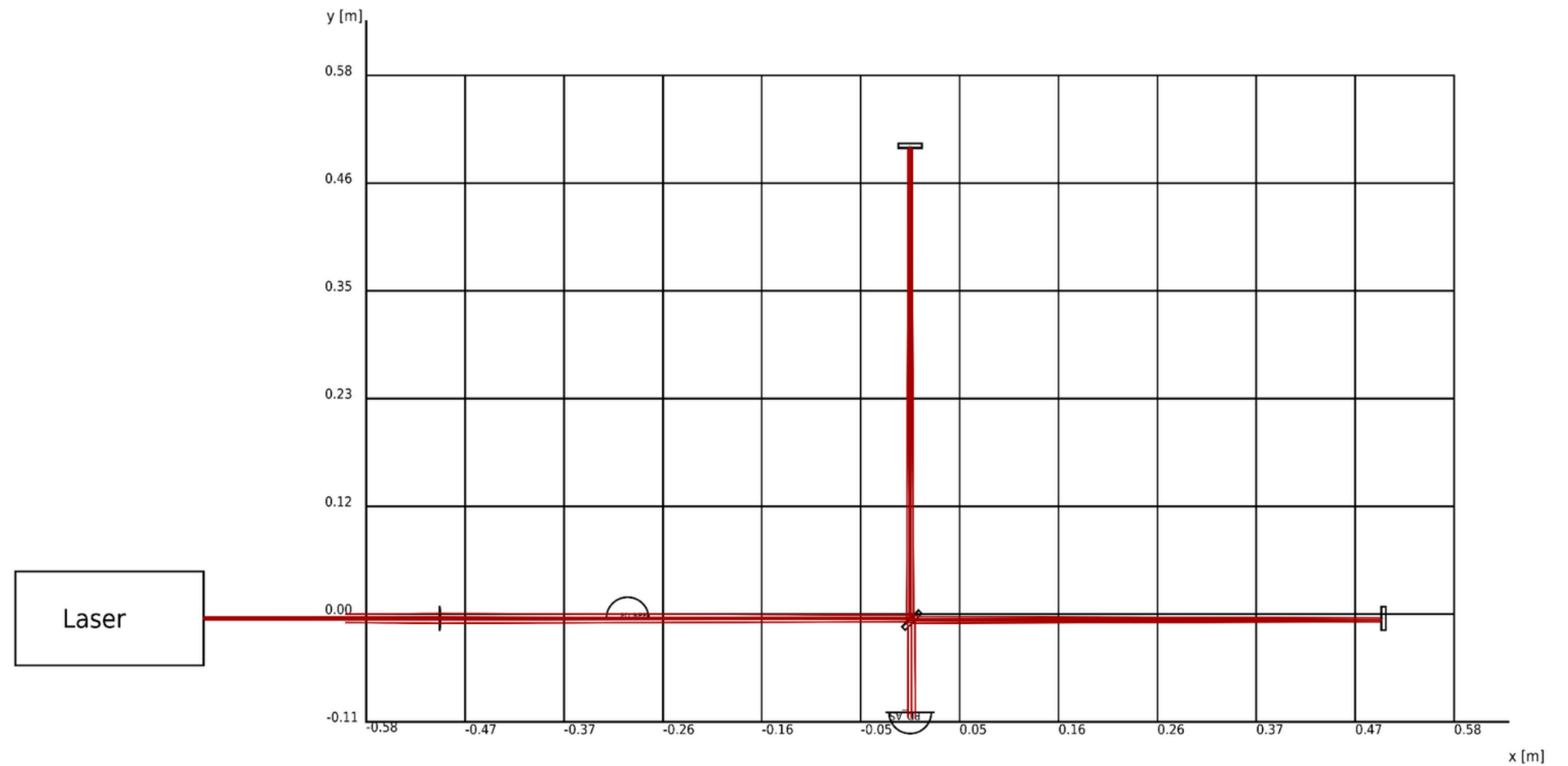


Optical Modeling Overview



Presented by: Kaylah McGowan
September 15th, 2026
LILA Core Group Meeting

LILA Design Considerations

Consideration	Implication for our model
Launch and deployment	Survive vibration and shock; recover alignment after deployment within mass, volume, and power limits
Lunar thermal environment	Model temperature-driven changes in spacing, curvature, and alignment separately from thermal noise
Vacuum and dust	Natural vacuum helps propagation; exposed optics still need contamination protection and scattered-light control
No suspensions	Calculate ground and support coupling explicitly, unlike LIGO’s suspended, isolated test masses
Remote operation	Include alignment adjustment, calibration, and monitoring with limited maintenance access

Preliminary Noise Study

No suspensions

Fixed optics: ground and support coupling still require modeling.

Notebook configuration: 1 km arms, 1064 nm, 1–100 mHz.

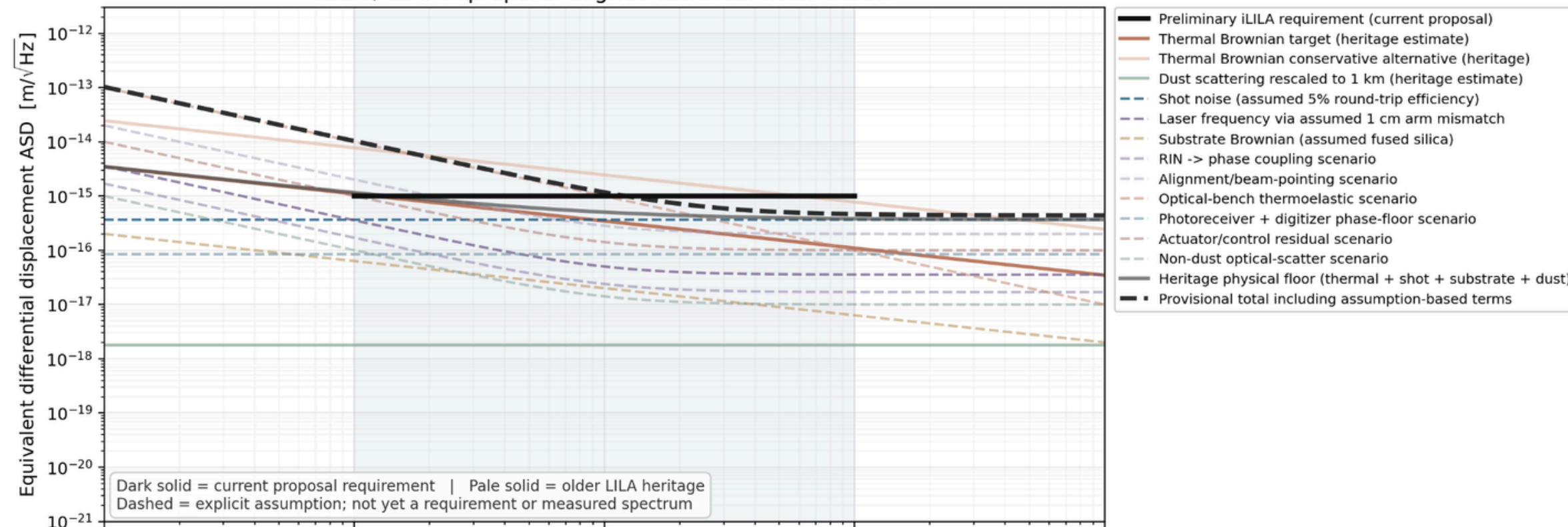
Preliminary target: 10^{-15} m/ $\sqrt{\text{Hz}}$.

Dashed curves are assumed scenarios, not measured performance.

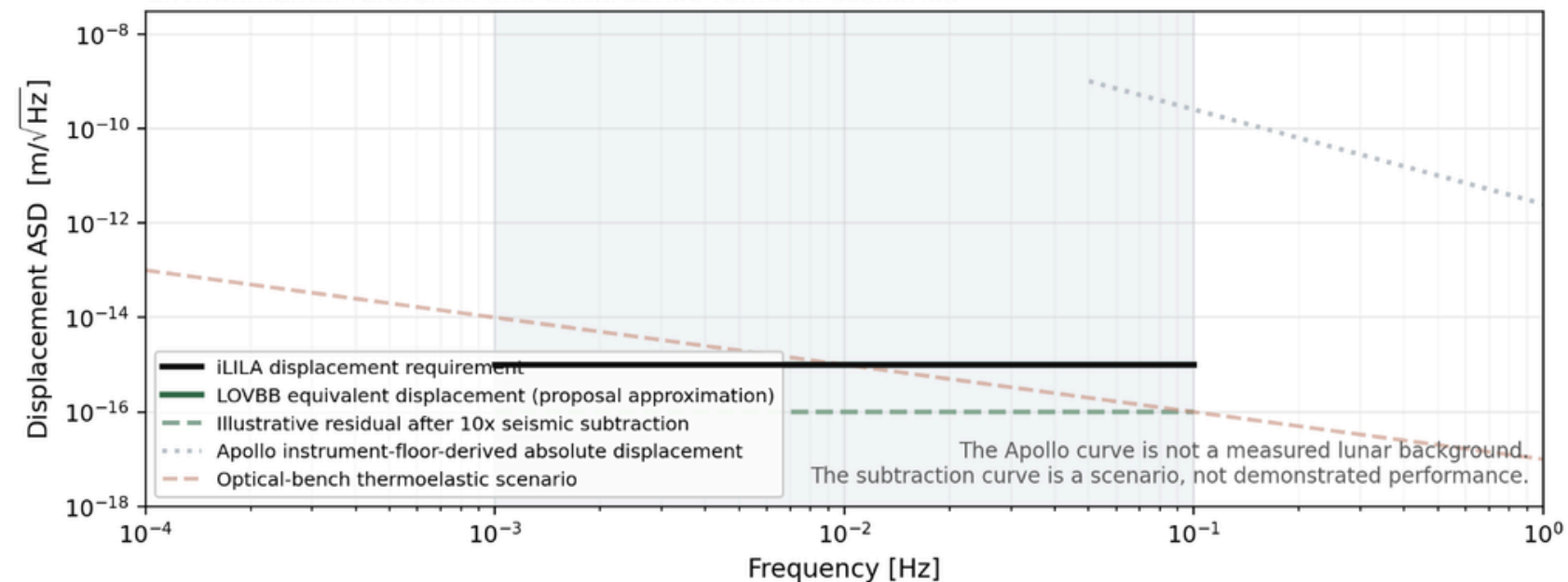
Separate from the 1550 nm bench rerun.

Current proposal: 1 km equal-arm dark-fringe interferometer | 1-100 mHz | $1\text{e-}15$ m/ $\sqrt{\text{Hz}}$

iLILA / LDC-IA proposal-aligned technical noise model



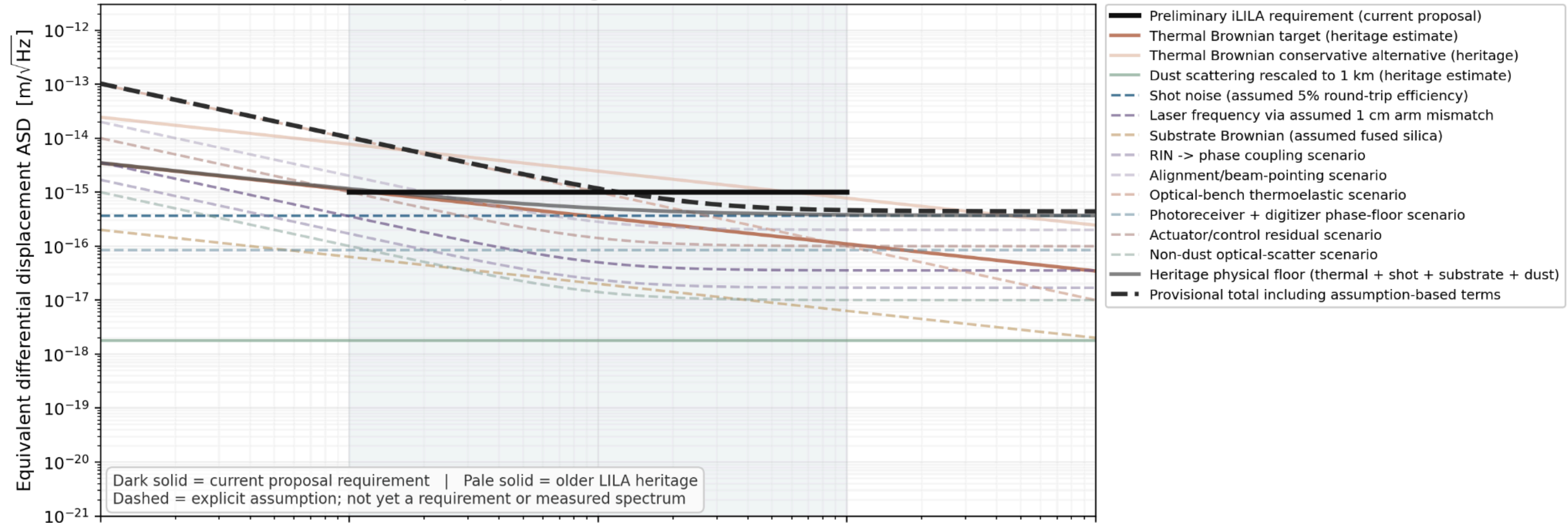
Environmental reference and disturbance-removal scenarios



Preliminary Noise Study

Current proposal: 1 km equal-arm dark-fringe interferometer | 1-100 mHz | $1e-15$ m/sqrt(Hz)

iLILA / LDC-IA proposal-aligned technical noise model



Implications

- Provisional noise exceeds the target below ~10 mHz.
- Thermal drift and pointing dominate the low-frequency estimate.
- Shot noise limits the higher-frequency estimate.
- Finesse and thermal modeling will test assumed couplings.

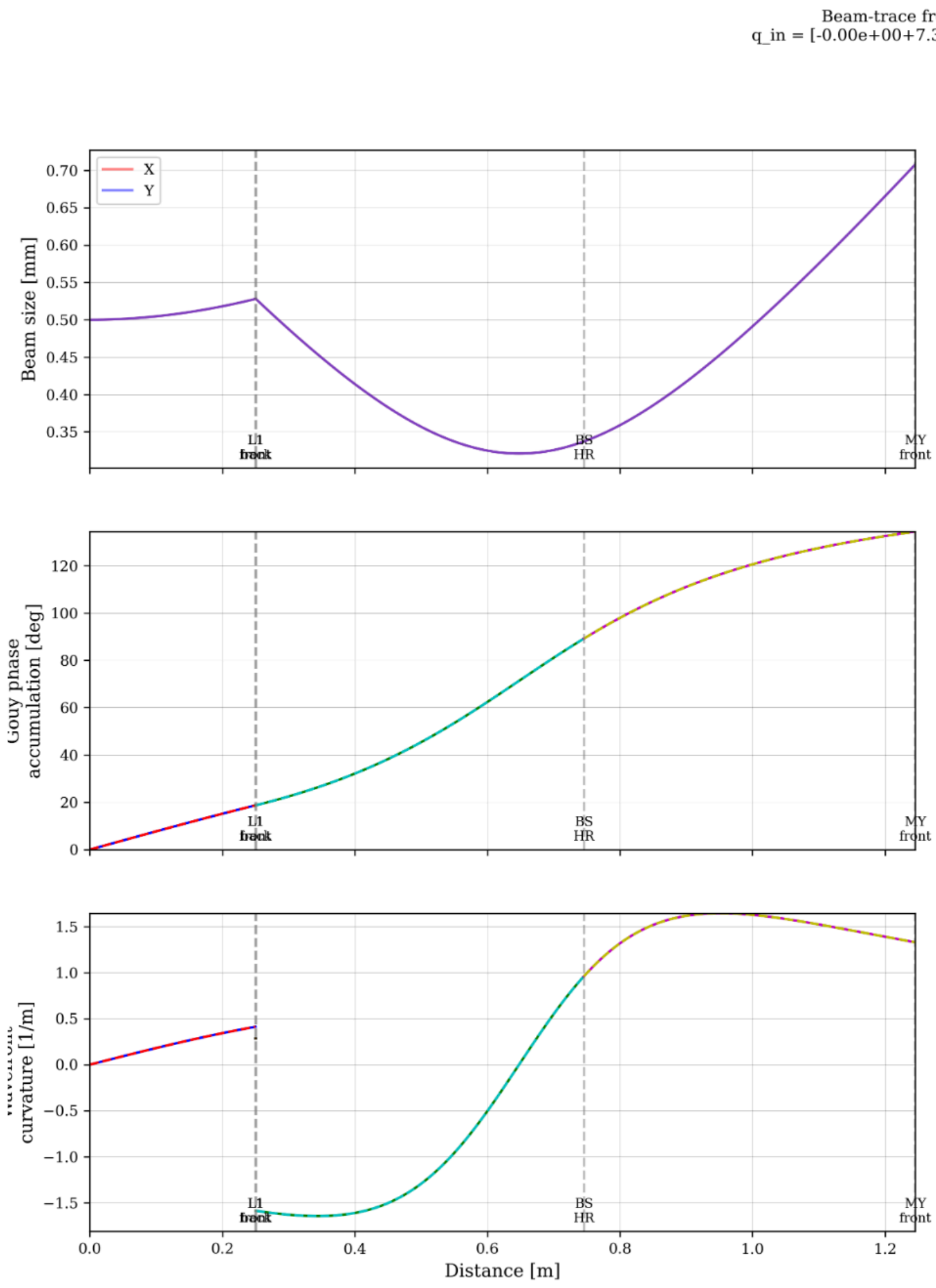
Current Fintrace Model

This slide presents a detailed view of the Fintrace model, showcasing the **laser, input lens, beamsplitter, and end mirrors**. It also illustrates the trace beam size, curvature, and Gouy phase along both arms, providing essential insights into optical performance.

Component	Parameter	YAML value
Laser	Power	1 W
Laser	Wavelength	1064 nm
Laser	Gaussian beam radius, w	0.5 mm
Laser	Wavefront radius of	∞ (planar)
Laser	Normal angle	0°
Input lens, L1	Focal length	0.50 m
Input lens, L1	Diameter	25 mm
Beamsplitter	Power transmission, T	0.5
Beamsplitter	Optical loss, L	0
Beamsplitter	AR-surface reflectivity,	0
Beamsplitter	Thickness	5 mm
Beamsplitter	Refractive index, nr	1.45
Beamsplitter	Front/back surface	∞ (planar)
Beamsplitter	Diameter	25 mm
Beamsplitter	Angle of incidence	45°
Beamsplitter	Normal angle	135°
End mirrors, MX and MY	Power transmission /	0 / 0
End mirrors, MX and MY	Surface radius of	∞ (planar)
End mirrors, MX and MY	Diameter	25 mm

Current Fintrace Model

This slide presents a detailed view of the Fintrace model, showcasing the **laser, input lens, beamsplitter, and end mirrors**. It also illustrates the trace beam size, curvature, and Gouy phase along both arms, providing essential insights into optical performance.

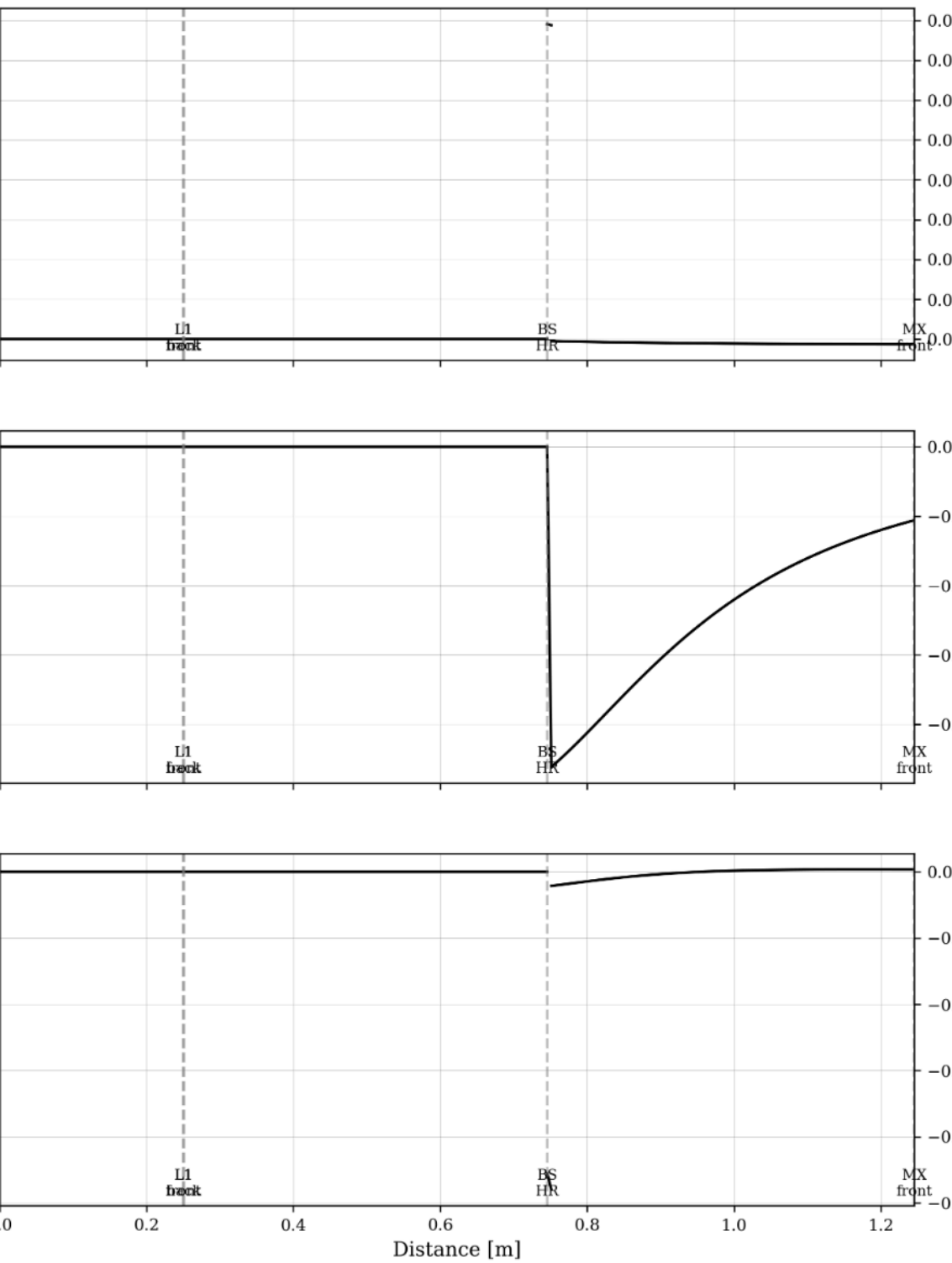
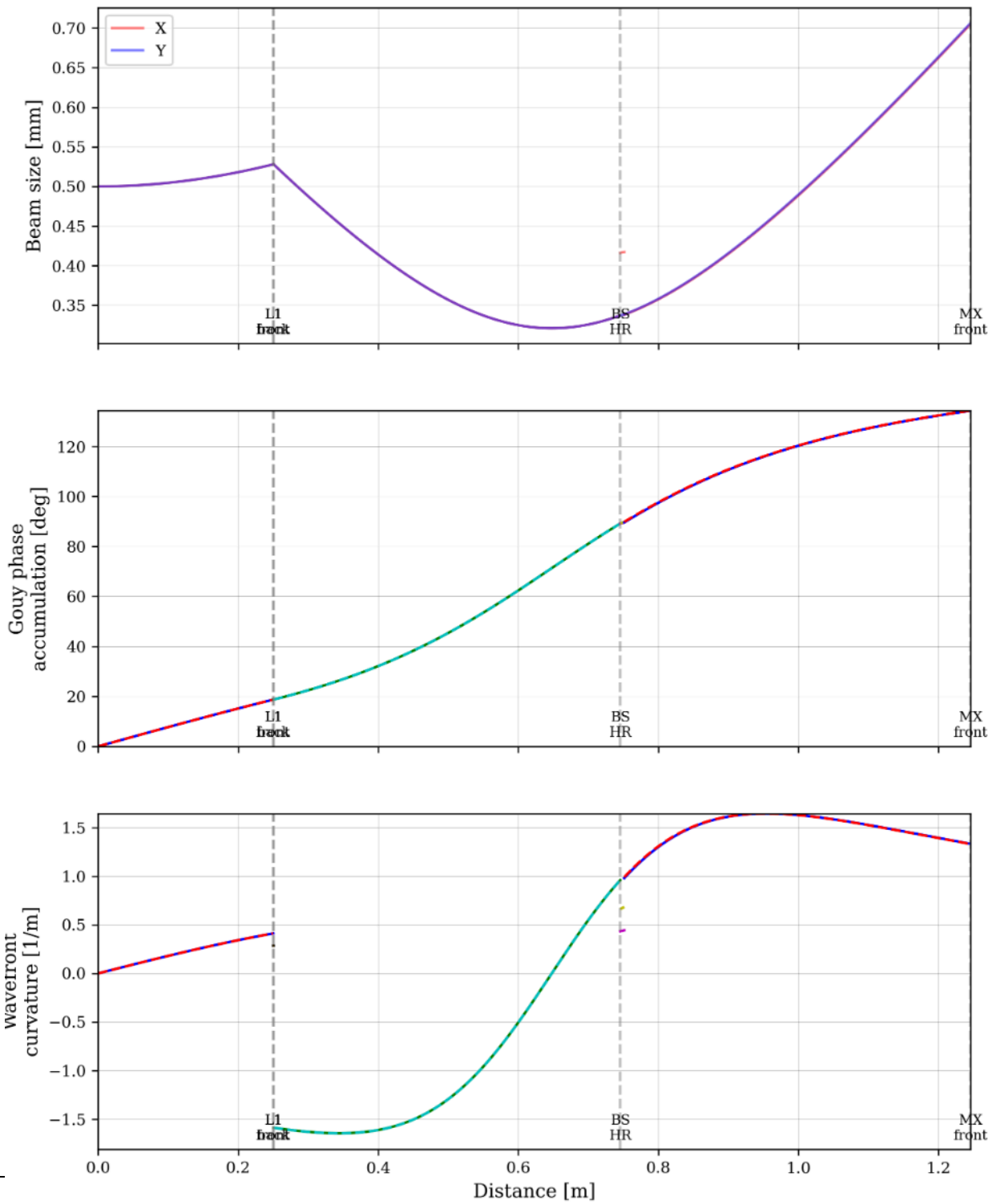


Gouy Phase Insights

Understanding its Role in Beam Dynamics

Gouy phase is crucial for determining **relative phases** between modes, aligning sensors effectively, and tracking beam characteristics.

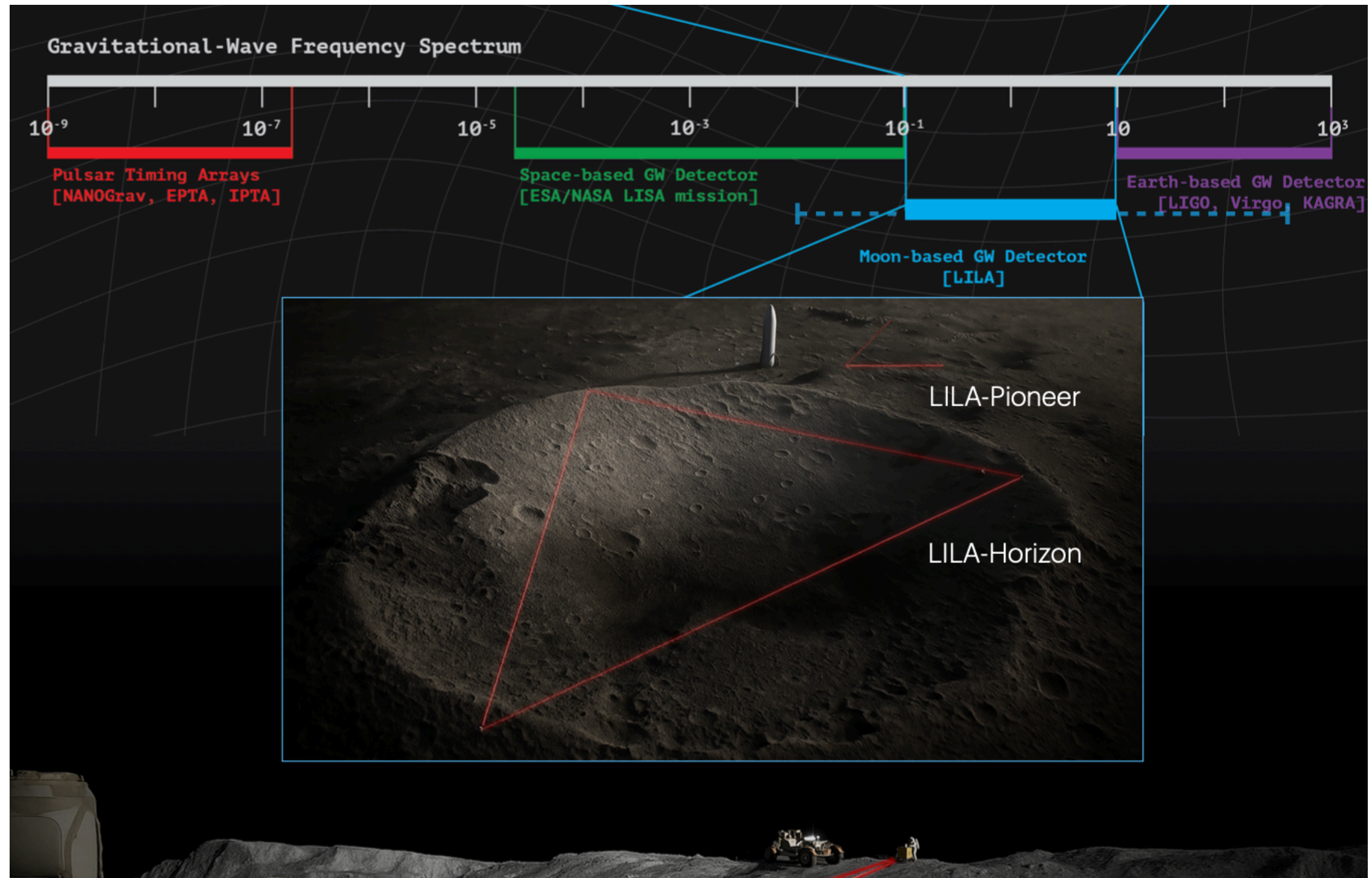
Beam-trace from Laser to MX_front
q_in = [-0.00e+00+7.38e-01j, -0.00e+00+7.38e-01j]



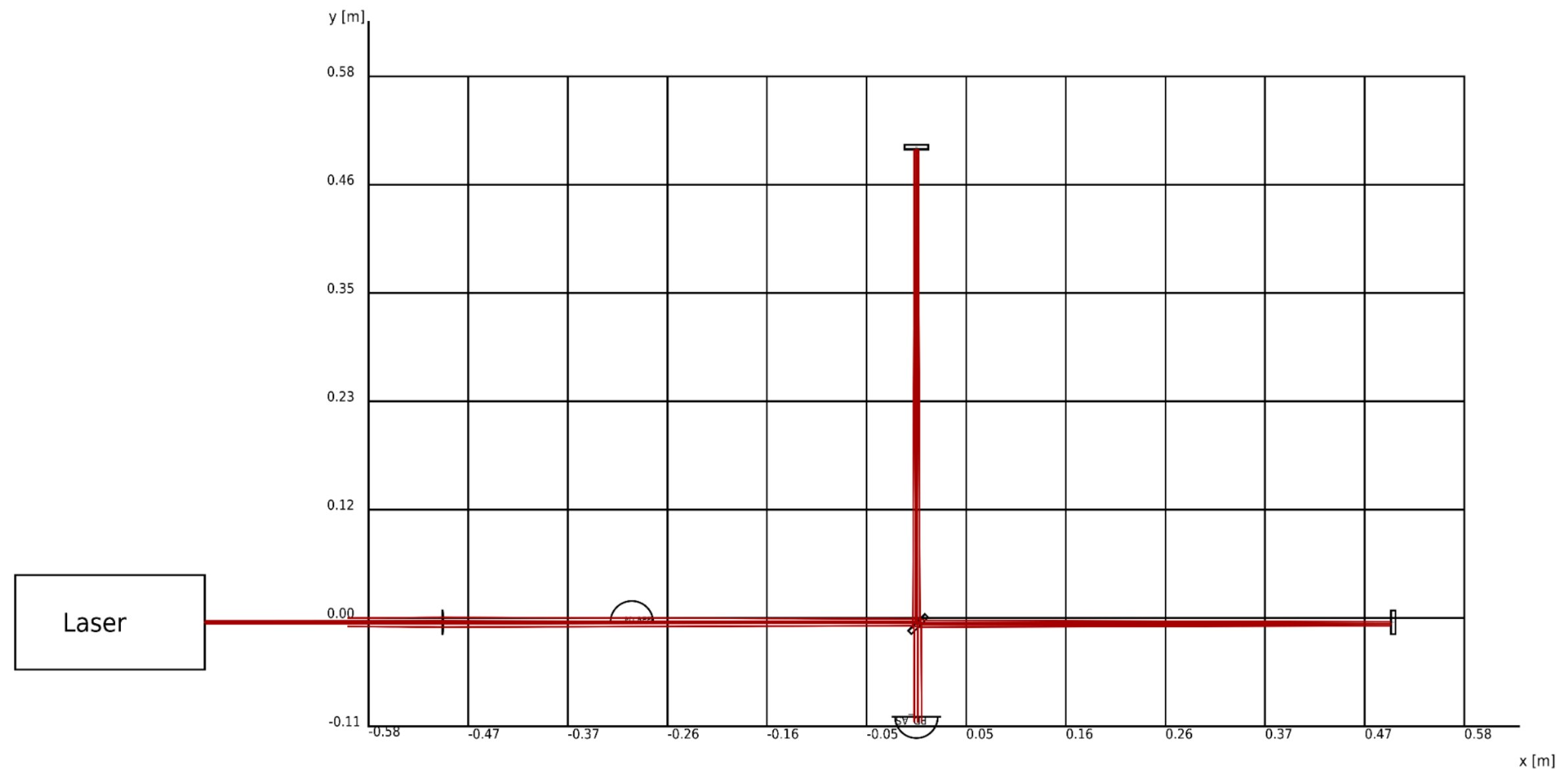
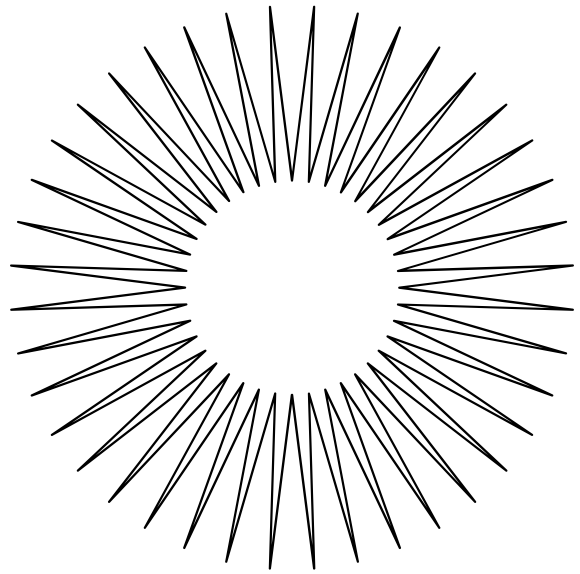
Beam Propagation

Analyzing Transverse Components in Two Arms

The MY and MX models illustrate **transverse overlap** and Gouy phase differences, essential for accurate alignment and beam performance.



Next Steps



1. Extend the optical model in Finesse
2. Incorporate the full layout and actual component parameters
3. Working on swapping out for the 1550 nm collimator and 1.25 mm beam radius
3. Study coating, substrate, and support thermal noise
4. Couple estimates to the readout and refine the LILA noise budget

Thank you! Suggestions & Feedback Appreciated